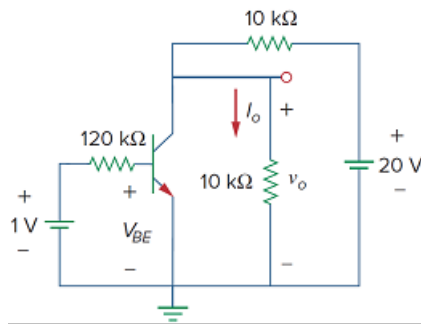


Problem

The transistor circuit in Fig. 3.45 has $\beta = 80$ and $V_{BE} = 0.7$ V. Find v_o and i_o .

Figure 3.45:



Step-by-step solution

Step 1 of 5

Refer to Figure 3.45 from the textbook for transistor circuit diagram.

The transistor has $\beta = 80$ and base to emitter voltage V_{BE} is 0.7 V. Use the mesh analysis to find the required values.

Replace the transistor with its equivalent circuit. The modified circuit with current loops is shown in Figure 1.

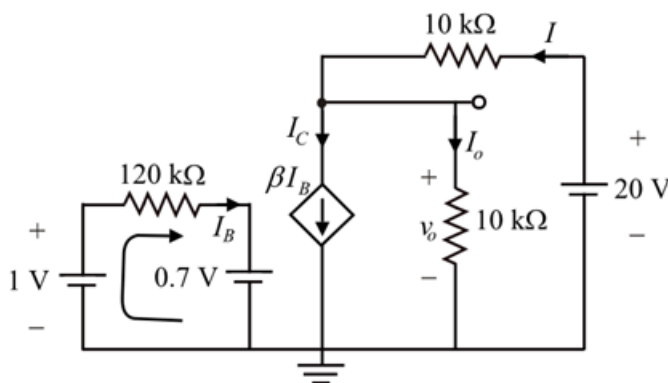


Figure 1

Step 2 of 5

Apply the Kirchhoff's voltage law loop I_B .

$$-1 + (120 \times 10^3) I_B + 0.7 = 0$$

$$(120 \times 10^3) I_B = 0.3$$

$$I_B = \frac{0.3}{120 \times 10^3}$$

$$I_B = 2.5 \mu\text{A}$$

Step 3 of 5

From Figure 1, observe that the collector current I_C is,

$$I_C = \beta I_B$$

Substitute the 80 for β and $2.5 \mu\text{A}$ for I_B .

$$\begin{aligned} I_C &= (80)(2.5 \times 10^{-6}) \\ &= 200 \mu\text{A} \\ &= 0.2 \text{ mA} \end{aligned}$$

Step 4 of 5

Apply Kirchhoff's current law at node v_o .

$$\begin{aligned} -I + I_C + I_o &= 0 \\ \frac{v_o - 20}{10 \times 10^3} + 0.2 \times 10^{-3} + \frac{v_o}{10 \times 10^3} &= 0 \quad (\text{since } I_C = \beta I_B) \\ \left(\frac{1}{10 \times 10^3} + \frac{1}{10 \times 10^3} \right) v_o &= 2 \times 10^{-3} - 0.2 \times 10^{-3} \\ 0.2 \times 10^{-3} v_o &= 1.8 \times 10^{-3} \\ v_o &= \frac{1.8 \times 10^{-3}}{0.2 \times 10^{-3}} \\ &= 9 \text{ V} \end{aligned}$$

Therefore, the voltage v_o is 9 V.

Step 5 of 5

The voltage v_o is the voltage across the resistor $10 \text{ k}\Omega$. The current through the $10 \text{ k}\Omega$ resistor is,

$$\begin{aligned} I_o &= \frac{9}{10 \times 10^3} \\ &= 900 \mu\text{A} \end{aligned}$$

Therefore, the current I_o is 900 μA .